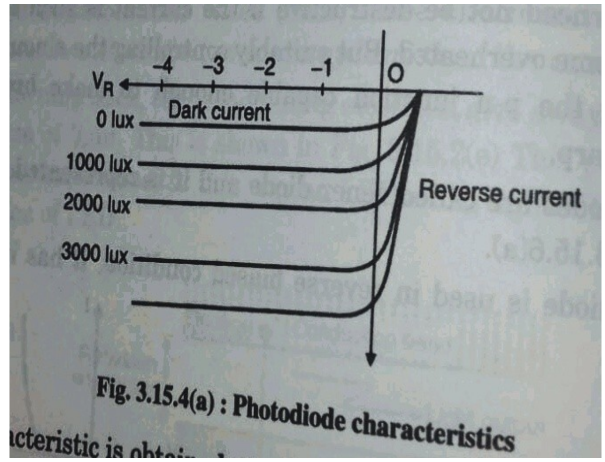
SEMICONDUCTOR

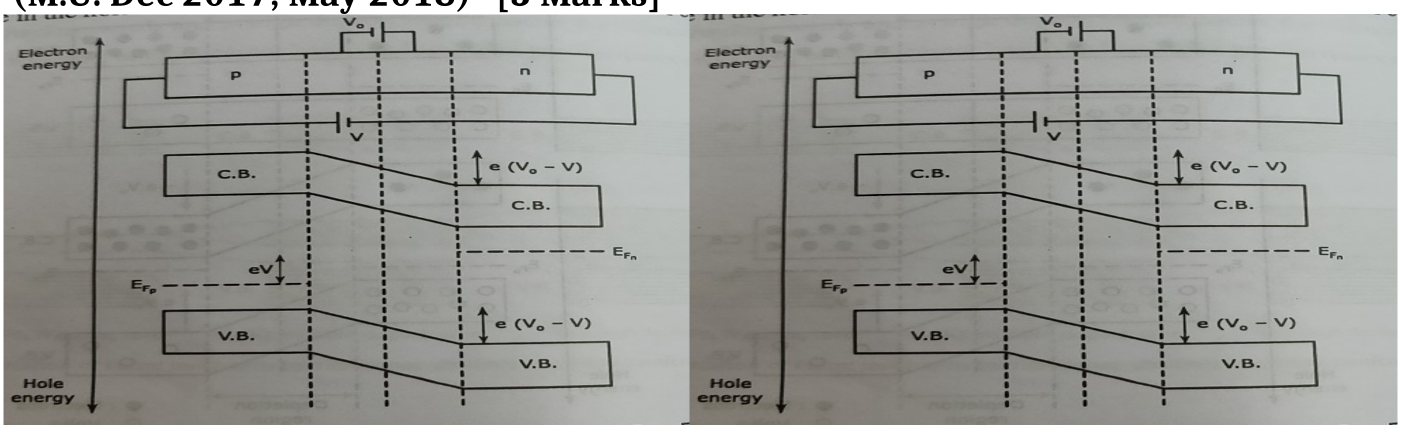
Q1. Draw the I-V characteristic of a photo diode. What is meant by dark current.
(M.U. Dec 2016) [3 Marks]



I-V characteristics of photo diode:

- Photo diode is always operated in reverse bias mode. Its I-V characteristic is as shown below:
- This characteristic is obtained at various values of illuminance (measured in lux)
- It is also quite evident that photo current is almost independent of applies reverse voltage. For zero luminance the photo current is almost zero except for small dark current.
- Dark current is the current through the diode for zero illumination. It will be non-zero due to back ground radiation and thermally excited minority saturation current.

Q2. Draw the energy band diagram of p-n junction diode in forward and reverse bias condition.
(M.U. Dec 2017; May 2018) [3 Marks]



Q3. What is the principle of solar cell. Write is advantages and disadvantages.
(M.U. May 2017) [3 Marks]

PRINCIPLE OF SOLAR CELL

A solar cell works on the principle of photovoltaic effect. It is a device that directly converts the energy in light into electrical energy.

ADVANTGES:

- Raw material i.e. amount of solar energy is available at no cost for operation of the solar cell, hence it is more useful in satellite communication.
- It is pollution free. It is not harmful to human life.
- It is most useful in remote areas, where traditional transmissions may be difficult.

DISADVANTAGES:

- Operation in the night is not possible.
- It has low efficiency and is presently not economical.
- It is a dilute source.

Q4. Explain the formation of Potential Barrier across the unbiased p-n junction region. **(M.U. Dec 2016) [5 Marks]**

In p-type of semiconductor, holes are majority carriers and its Fermi level is just above the top of acceptor level at room temperature (300^0K). For n-type of semiconductor, electrons are majority carriers and its Fermi level is just below the donor level at 300^0K

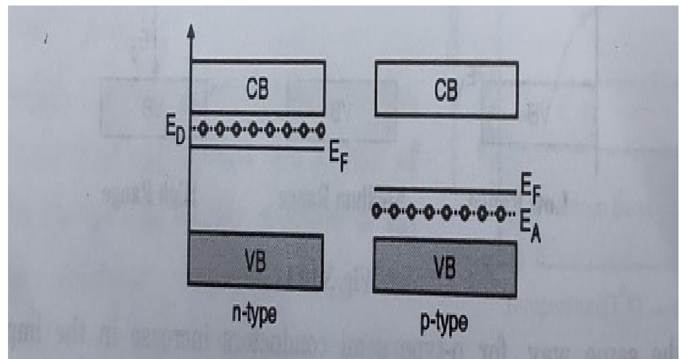


Fig. 3.11.1 : Position of Fermi-level in n-type and p-type semiconductor

- When p-n junction is being formed, both materials will find their Fermi levels non-aligned as shown in (fig a) and they must be aligned.
- In order to obtain alignment, holes from p-type (majority carriers for a p-type) move towards n-type and at the same time electrons from n-type (majority carriers for a p-type) move towards n-type and at the same time electrons from n-type (majority carriers from n-type) move towards p-type. This is called diffusion.

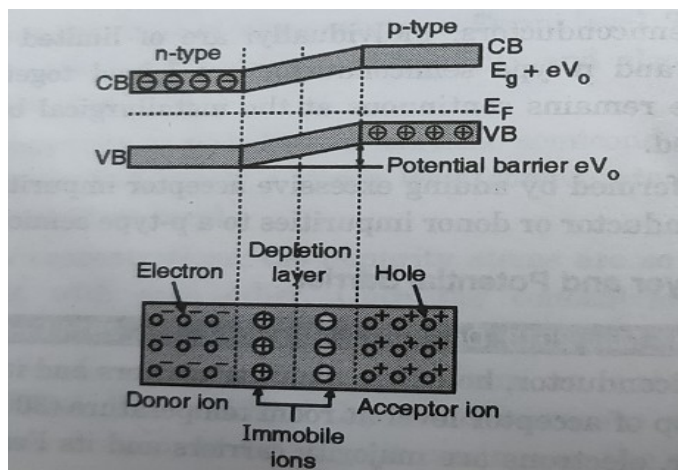


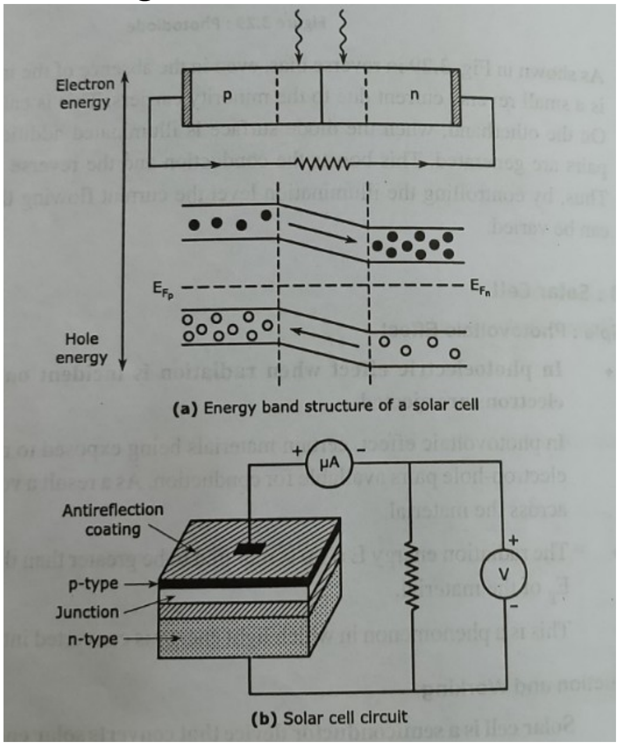
Fig. 3.11.2 : Unbiased p-n junction

- As the diffusion takes place across the junction the holes and the electrons combine to neutralize each other at the junction and produce a free space called depletion layer as shown in (fig b)
- At the junction due to the migration of a few holes from p-type to n-type region, the negative immobile ions are produced in p-region and in the same way positive immobile ions are produced in n-region as shown in (fig b)
- As the name suggest these ions are immobile in nature (fixed in lattice), under equilibrium condition it can be assured similar to a charged parallel plate capacitor and creates an electronic potential called potential barrier or junction potential which prevents further diffusion of majority charge carriers.

Q5. What is photovoltaic effect? Explain the principle and working of Solar cell.
(M.U. Dec 2012, 2013, 2017, 2016; May 2013, 2014, 2015, 2016, 2017, 2018) [5 Marks]

A. PHOTOVOLTIC EFFECT

When a suitable light falling on a p-n junction produces a potential difference across it. This voltage is capable of driving a current through an external circuit; this important phenomenon is called the "photovoltaic effect"



(a) Energy band structure of a solar cell

(b) Solar cell circuit

B. SOLAR CELL

a. PRINCIPLE

- A solar cell works on the principle of photovoltaic effect. It is a device that directly converts the energy in light into electrical energy
- It is first developed by French physicist Antoine-Cesar Becquerel, he observed the photovoltaic effect while experimenting with a solid electrode in an electrolyte solution when he saw a voltage developed when light fall upon the electrode.

b. CONSTRUCTION AND WORKING

- Solar cell is a semiconductor device that converts solar energy into electrical energy

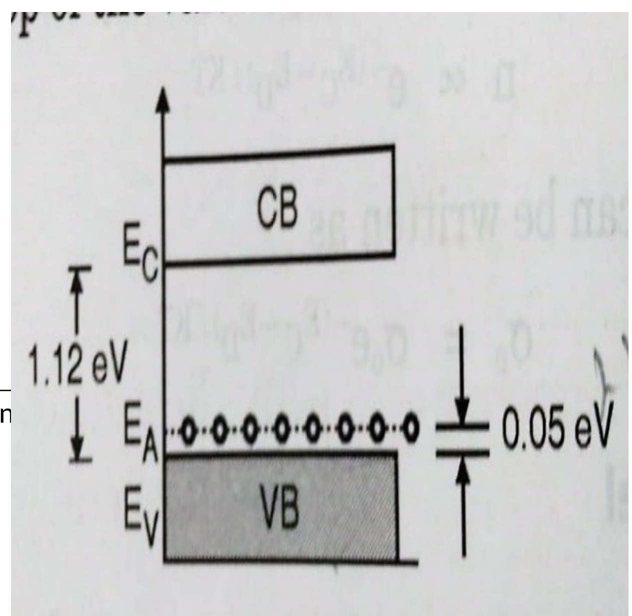
- This is a p-n junction diode with very high doping level. Solar cells have a flat shape with a very thin top layer. So that the incident solar energy can reach the junction area
- As the solar radiation is incident on the device, due to the radiation energy $E=h\nu \geq E_g$ electrons hole pairs are generated in p and n both the regions
- In the energy band structure of the solar cell (fig a)it is seen that the conduction band is lower in the n region than that in p region. Hence, the generated electrons of the conduction band of p-region travel to the conduction band of n-region which is at a lower electron energy level. Similarly, the holes created in the valence band of the n-region move to the valence band of the p region at the lower hole energy level
- The diffusion of electrons and holes through the junction constitutes the current
- The top surface of the solar cell is coated with an antireflection film to maximize the utilization of the incident solar energy by the junction as shown in (fig b)
- As a result, the diffusion charge carriers are accumulated on both sides of the junction viz. The electrons on the n side and holes on the p side. This results in a potential difference called the photo emf across the junction. This photo emf is proportional to the intensity of the incident radiation
- The photo emf can be measured and can be collected as the output by connecting a load resistance R_L across the cell
- However, a single solar cell can produce a photo voltage on only 0.5 – 0.6 V for Si and 0.1 V for Ge. That is why several solar cells are connected in series to generate the required voltage
- A solar cell does not need a power supply. It generates power
- Materials used for solar cells are different types of semiconductors, single crystal, polycrystal, thin silicon wafers, etc.

Q6. With energy band diagram, explain the variation of Fermi energy level with impurity concentration in extrinsic semiconductor.

(M.U. Dec 2010, 2014, 2016, 2017, 2018; May 2013, 2017, 2018) [5 Marks]

The position of fermi level is also affected by addition of impurity and also by variation in the concentration of impurity.

- If a donor impurity is added to an intrinsic semiconductor, it results in n-type



semiconductor and a donor level comes into existence below bottom of conduction band.

- At impurity concentrations, impurity atoms are so spaced so that they do not interact with each other. Once the concentrations are increased, interaction among them starts.
- The donor levels splitting and form an energy band below conduction band. The width of this band increases with increase in impurity concentration. At one stage it overlaps with the conduction band.
- Due to broadening donor levels into band reduces the width of forbidden energy gap and Fermi level is found moved upwards.
- With increase in concentration of donor impurity the Fermi level continues shifting towards conduction band and enters into conduction band.
- In the same way for p-type semiconductors increase in the impurity concentration makes Fermi level shift into valence band.

Q7. Write Fermi Dirac Distribution Function. With the help of diagram explain variation of Fermi Level with Temperature in n-type semiconductor.

(M.U. Dec 2009, 2010, 2012, 2013, 2015, 2016, 2017, 2018; May 2013, 2017, 2019) [5 Marks]

According to Fermi- Dirac Statistics, we make use of function f(E) which determines the carrier occupancy of the energy. In other words, f(E) governess the distribution of electrons among the energy levels as a function of temperature. It is given by

$$F(E)= \frac{1}{1 + \exp(E - E_f) / KT}$$

Where f(E)= Probability that a particular energy level E is occupied by an electron.

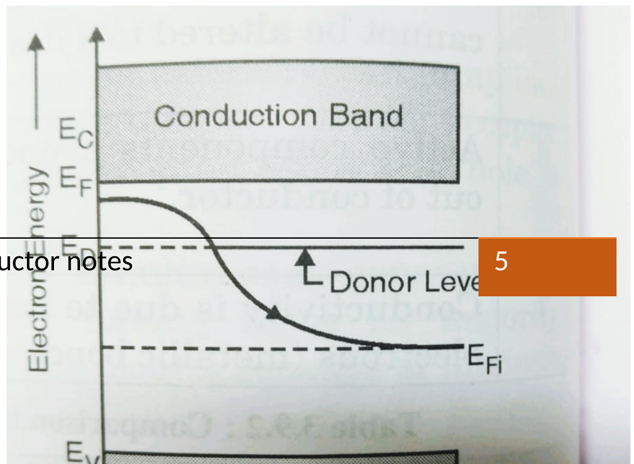
E_f=Fermi energy

K=Boltzmann constant

T=Temperature in Kelvin

Effect of Temperature on n-Type Material

- At lower temperature- When the temperature in the semiconductor is low, only few donor atoms get ionized and electrons move from the donor level to the conduction band. Hence Fermi level



for n-type semiconductor at low temperature lies midway between the bottom of the conduction band and donor level.

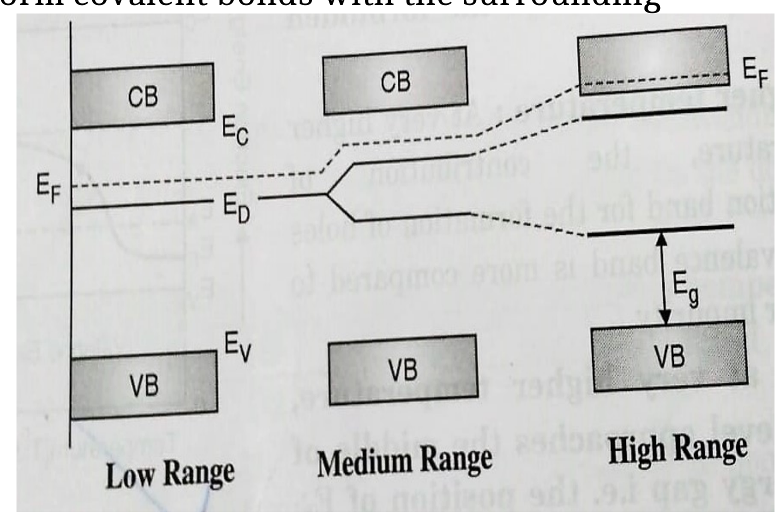
- At Moderate Temperature: At moderate temperature all donor atoms are ionized. So, the concentration of electrons in conduction band is equal to the concentration of donor atoms. When the temperature increases up to moderate value, Fermi Level slowly shifts away from the conduction band and moves towards the center of the forbidden gap.
- At higher temperature: At high temperature, the concentration of transfer of electrons from valence band to conduction band is more compared to concentration of electrons from donor atoms and Fermi Level is shifter to middle of forbidden gap.

The variation of Fermi level with temperature for n-type of material is shown in figure.

Q8. How does the position of Fermi energy level changes with increasing doping concentration in p-type semi-conductors? Sketch the diagram.

(M.U. Dec 2014, 2016, 2018; May 2013, 2017, 2018) [5 Marks]

If a trivalent impurity is added to a pure semiconductor it becomes a p-type extrinsic semiconductor. Each impurity atom tries to form covalent bonds with the surrounding other atoms and fall short of one electron for completing four covalent bonds.



- As a result a vacancy is left in the bonding. This vacancy is not a hole.
- The addition of nan impurity adds an allowed level E_A at a very small distance above the top of valence bond as shown in the figure.
- At T=0K all acceptor levels are vacant and valence band is full while conduction band is empty.
- When the temperature increases, electrons from the valence band jump into acceptor level and leave behind holes.
- The generation of holes is not followed by simultaneous generation of electrons. Therefore, p-type semiconductor has holes as majority carriers.
- Hence in p-type semiconductor as there are any free holes in valence band and due to impurity concentration, the Fermi level gets shifted towards the valence band. At 0K it is between top of valence band and the level E_A.

Q9. Explain the concept of Fermi level. Prove that the Fermi level is exactly at the center of the Forbidden energy gap in intrinsic semiconductor.

(M.U. Dec 2009, 2010, 2012, 2013, 2016, 2017; May 2013, 2017, 2019) [7 Marks]

It can be shown for intrinsic semiconductors, Fermi energy level E_F lies mid-way between conduction and valence band. The proof is given below

At any temperature T 0^0K,
n_e Number of electrons in conduction band
n_v Number of holes in valence band

We have $n_e = N_C e^{-(E_C - E_F)/KT}$

Where Nc= Effective density of states in conduction band

And $n_v = N_V e^{-(E_F - E_V)/KT}$

Where N_v=effective density of states in valence band

For best approximation $N_C = N_V$

For intrinsic semiconductor

$N_c = N_v$ (Densities are approximately equal)

$N_c \cdot e^{-(E_C - E_F)/KT} = N_V \cdot e^{-(E_F - E_V)/KT}$

$$\frac{e^{-(E_C - E_F)/KT}}{e^{-(E_F - E_V)/KT}} = \frac{N_V}{N_C}$$

$$e^{-(E_C - E_F - E_F + E_V)/KT} = \frac{N_V}{N_C}$$

$$e^{-(E_C + E_V - 2E_F)/KT} = \frac{N_V}{N_C}$$

As $N_C = N_V = 1$

$$e^{-(E_C + E_V - 2E_F)/KT} = 1$$

$\therefore$ Taking ln on both side

- = 0

$\therefore E_C + E_V = 2E_F$

$$\therefore \quad E_F = \frac{(E_C + E_V)}{2}$$

Thus, the Fermi level in an intrinsic semiconductor lies at the middle of forbidden energy gap.

Q10. State the Hall effect. Derive the expression for Hall Voltage and Hall coefficient with neat diagram.

(M.U. Dec 2008, 2009, 2011, 2012, 2013, 2016, 2017; May 2011, 2013, 2018) [7/8 Marks]

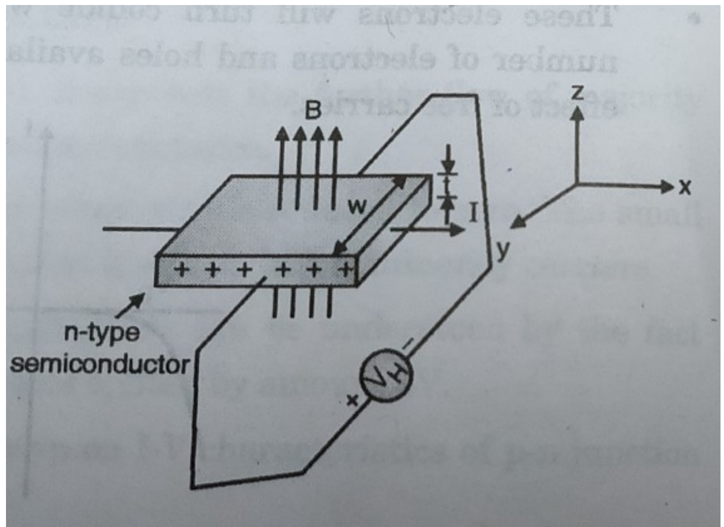


Fig. 3.14.1 Hall effect

- If a metal or semiconductor, carrying a current I is placed in a transverse magnetic field B, an electric field E is induced in the direction perpendicular to both I and B. This phenomenon is known as Hall effect and the electric field or voltage induced is called Hall voltage (V). The physical process of the hall effect is as follows. Consider a specimen along positive x-direction (fig)
- The current flowing through the specimen is in the positive x-direction and the magnetic field is in the positive z-direction. The force exerted on charge carriers that is, on electrons is downward. The electrons move downward and thus voltage H (Hall voltage) is developed with upper surface as positive and lower as negative Fig a is as follows
- If the specimen in fig a is assumed to be of n-type, then magnetic force experienced by electrons will be towards Y- direction as it is applied transversely

  $\therefore$ Magnetic force (F) = evB

- Holes present in specimen will experience the same force but in positive Y-direction. Hence electrons and holes will be separated
- This develops potential difference between both the surface denoted by V_H called hall voltage

$$\therefore\ E_H = \frac{V_H}{w} \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

- The current in case is given by

$$I = nAev \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

- In equilibrium condition, the force due to magnetic field b and the force due to electric field E_H Acting on the charge are balanced

$$\therefore\ eE_H = evB$$

Or $E_h = vB$ ....(3)

Using equation (1) and (3)

$$V_H = Bvw$$

Using equation (1) and (2)

$$v = \frac{I}{enA} = \frac{J}{en} \qquad \text{where, } J = \frac{I}{A}$$

Hence hall voltage can be written as

$$V_H = \frac{IBw}{enA} = \frac{Bw}{en} \times J$$

It can be modified by using, $A = w \times t$

$$V_H = \frac{IB}{ent} \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(4)$$

By measuring V_H, I, B, and t the charge density ($n_h e$ or $n_e e$) can be calculated

Another important parameter is hall coefficient R_H it is defined as

$$R_H = \frac{1}{pe} \quad \text{(for p-type semiconductor)}$$

$$R_H = \frac{1}{ne} \quad \text{(for n-type semiconductor)}$$

From equation 4

$$R_H = \frac{V_H t}{BI} \qquad \text{.....................(5)}$$

Q11. State applications of Hall Effect.

(M.U. Dec 2008, 2009, 2011, 2012, 2013, 2015, 2016, 2017; May 2011, 2016, 2017, 2019) [3/7/8 Marks]

APPLICATIONS OF HALL EFFECT

- The all voltage V_H is proportional to magnetic field B , for the given current I, therefore hall effect is used in magnetic field meter.
- The charge carrier concentration can be determined.
- The mobility of charge carriers can be determined.
- The nature of semiconductor can be determined.

Q12. Define the terms drift velocity, diffusion current, P-N junction and mobility of charge carriers.

(M.U. Dec 2011, 2012, 2016; May 2010, 2017) [4/8 Marks]

Drift velocity-When an electric field is applied though the randomness in its motion persists there is overall shift in its position in the direction of the field with time. It means a net transportation of charge resulting in a current in the direction of the field. The net displacement in the electrons position per unit time caused by the application of the electrons in the steady state in an applied electric field is called the drift velocity.

Diffusion current- Non uniform concentration of charge carriers produce diffusion current. This kind of non-uniform concentration of charge carriers can be formed by thermal or radiation excitation of a part of the material or by injecting carriers in to the material through surface. Whenever concentration of electrons or holes due to excitation is noticed these excess carriers diffuses to low concentrated place.

The rate of diffusion is proportional to concentration gradient and forms diffusion current. In a semiconductor, total current is due to drift and diffusion currents.

P-N junction: The extrinsic semiconductor, individually are of limited use. However, when a p-type and n-type semiconductor are joined together such that crystal structure remains continuous at the metallurgical boundary, a p-n junction is formed. In practice it is formed by adding excessive acceptor impurities to a portion of n-type semiconductor or donor impurities to a p-type semiconductor.

Mobility

The drift velocity is linearly proportional to the electric field intensity

Thus, $V_d \alpha \varepsilon$

$V_d = \mu\varepsilon$ where, the symbol ε is used for electric field intensity

The constant of proportionality is called mobility of charge carriers. It indicates how fluently a charge carrier can sweep through electric field. It can be defined as the drift velocity acquired per unit electric field intensity. Its S.I. unit is m^2/v-sec.